AI Learning Navigator – Student Performance Analysis System

Submitted by:

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1. OBJECTIVE:

The objective of this project is to develop an AI-driven system that personalizes learning for students based on their academic performance, attendance, and preparation levels.

The system aims to move beyond traditional one-size-fits-all education by analyzing student data and generating personalized learning paths, recommendations, and career alignment insights.

2. PROBLEM STATEMENT:

Educational institutions manage diverse student populations with varying learning speeds, strengths, and career aspirations.

Traditional education systems often follow standardized curricula, which:

• Fail to adapt to individual learning needs
• Overlook different cognitive and learning styles
• Do not align strongly with evolving job market demands

Educators and academic planners require deeper insights into:
• Identifying students who are at risk of falling behind or disengaging

• Adapting learning pace and content difficulty for individual students
• Recognizing skills relevant for future career success
• Personalizing learning journeys while maintaining curriculum standards

The goal of this project is to build an AI-driven system that dynamically generates personalized learning paths, improves student performance, and enhances future career readiness.

3. METHODOLOGY:

The project follows a structured data-driven approach:

1. **Data Preprocessing:**

   - Handled missing values in test preparation
   - Standardized categorical data

2. **Exploratory Data Analysis (EDA):**

   - Study time vs performance analysis
   - Absence impact analysis
   - Test preparation comparison
   - Correlation heatmap

3. **Clustering:**

- Applied KMeans clustering
- Segmented students into:
  - Low Performer
  - Average (At Risk)
  - High Performer

4. **Recommendation System:**

- Rule-based logic using multiple conditions
- Generated personalized suggestions

5. **Skill Gap Analysis:**

- Identified student learning levels:
  - Weak Fundamentals
  - Needs Improvement
  - Strong

6. **Learning Path Generation:**

- Personalized paths based on performance level

7. **Career Alignment:**

- Suggested career directions based on performance and consistency

8. **Action Plan:**

- Assigned priority levels:
  - High Priority
  - Medium Priority
  - Low Priority

4. FLOW DIAGRAM:

5. SAMPLE INPUT:

math_score, absences, study_time_weekly, test_preparation

6. SAMPLE OUTPUT:

math score	performance_ level	skill_ gap	learning_path	career_ path	action_ plan
99	High Performer	Strong	Advanced learning	Data Science	Low Priority
86	Average (At Risk)	Strong	Improve consistency	Needs consistency	Medium Priority
83	High Performer	Strong	Skill Development	Focus on Foundation	Low Priority

7. Dashboard Implementation:

An interactive dashboard was developed using Streamlit to visualize and analyze student performance in real-time.

The dashboard provides a user-friendly interface with dynamic filters and multiple visualizations, enabling users to explore insights effectively.

Key Features:

• Interactive filters for performance level, gender, course, score range, and absences

• Real-time data updates based on user selection

• Visualization of student distribution and category breakdown

• Scatter plots for study time vs performance and absences vs performance

• Box plot for test preparation impact

• Correlation heatmap for multi-variable analysis

• Identification of top-performing and at-risk students

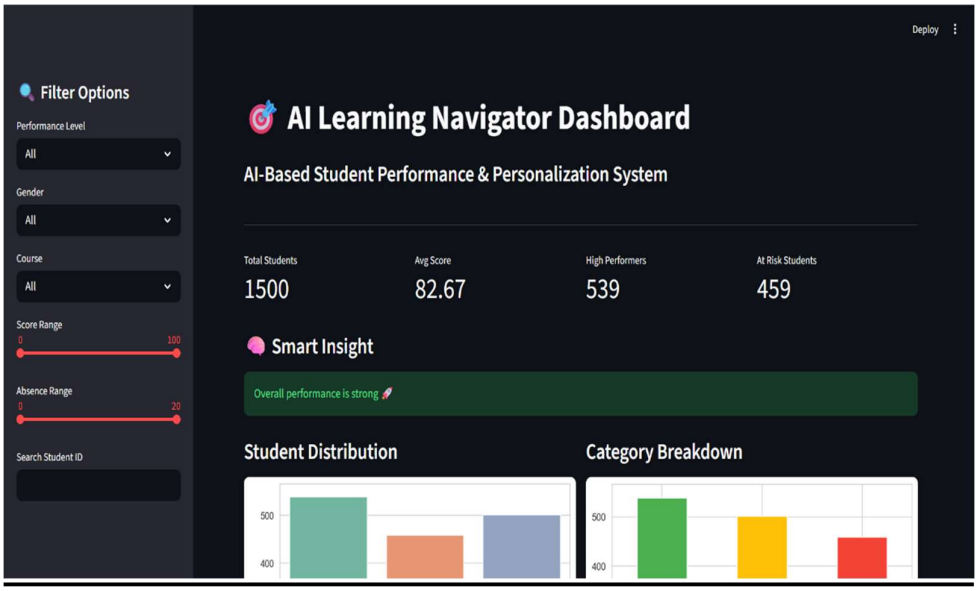
• Downloadable dataset for further analysis

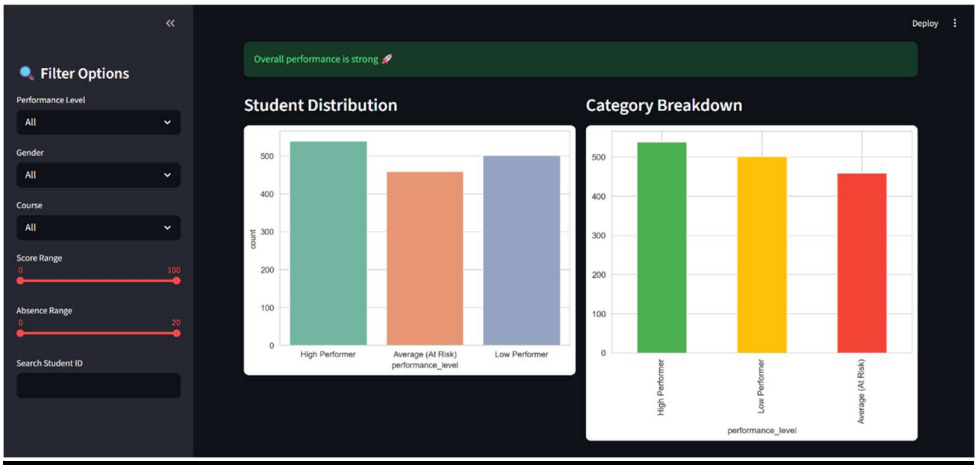
Purpose:

The dashboard enhances decision-making by transforming static analysis into an interactive system. It allows educators to identify patterns, monitor student performance, and take data-driven actions.

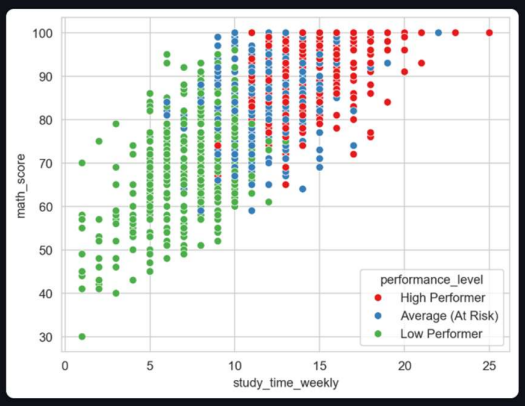
This integration demonstrates the practical application of AI and data analytics in building scalable educational solutions.

Screenshots:

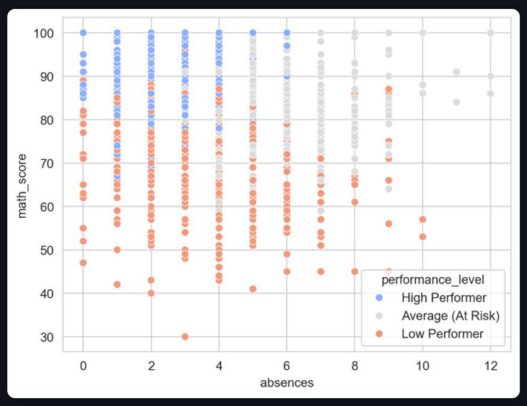




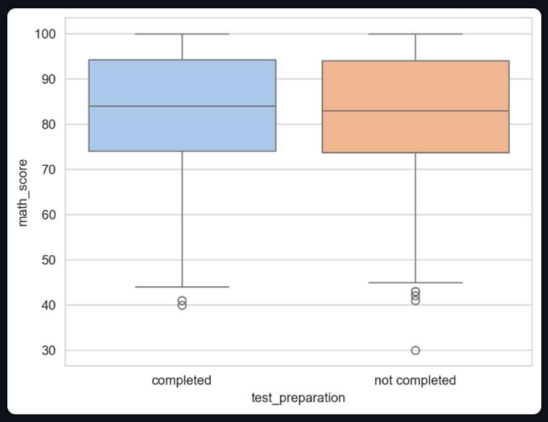
Study Time vs Score



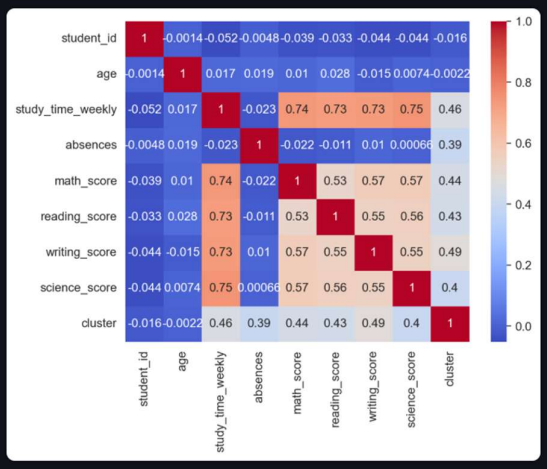
Absences vs Score



Test Preparation Impact



Correlation Heatmap



🏆 Top Performers

	student_id	course_id	age	gender	ethnicity	study_time_weekly	absences	test_preparation	math_score	reading_score	writing_score	science_score
5	6	C102	20	Female	Group B	16	9	completed	100	94	76	90
384	385	C105	17	Female	Group A	21	3	not completed	100	100	100	100
385	386	C101	17	Female	Group D	12	3	completed	100	86	97	80
330	331	C103	19	Male	Group E	10	1	completed	100	83	74	85
331	332	C102	18	Female	Group E	15	2	completed	100	87	83	77
389	390	C102	18	Male	Group D	17	6	completed	100	91	100	90
391	392	C105	19	Female	Group C	20	4	not completed	100	100	100	100
394	395	C105	17	Male	Group E	14	1	completed	100	88	98	98
396	397	C105	21	Male	Group B	19	7	completed	100	95	100	96
403	404	C105	17	Female	Group C	13	7	not completed	100	100	100	100

⚠️ At Risk Students

	student_id	course_id	age	gender	ethnicity	study_time_weekly	absences	test_preparation	math_score	reading_score	writing_score	science_score
1	2	C105	18	Male	Group B	13	12	completed	86	100	68	85
5	6	C102	20	Female	Group B	16	9	completed	100	94	76	90
8	9	C103	17	Female	Group D	8	7	not completed	71	83	76	63
12	13	C105	18	Male	Group B	10	4	not completed	83	93	98	66
13	14	C102	21	Male	Group A	14	5	not completed	64	82	100	99
14	15	C104	20	Female	Group C	11	9	completed	90	96	82	83
19	20	C104	21	Male	Group E	10	6	not completed	85	81	71	96
21	22	C105	21	Male	Group A	17	7	completed	100	82	100	100
27	28	C102	18	Female	Group A	14	6	completed	90	89	100	90
29	30	C104	18	Female	Group C	13	5	completed	99	87	88	100

Deploy

Filter Options

Performance Level

All

Gender

All

Course

All

Score Range

0 100

Absence Range

0 20

Search Student ID

Student Insights

	math_score	performance_level	skill_gap	learning_path	career_path	action_plan
0	99	High Performer	Strong	Advanced learning	Data Science / Engineering	Low Priority
1	86	Average (At Risk)	Strong	Improve consistency	Needs consistency	Medium Priority
2	83	High Performer	Strong	Advanced learning	Analytical roles	Low Priority
3	96	High Performer	Strong	Advanced learning	Data Science / Engineering	Low Priority
4	89	High Performer	Strong	Advanced learning	Data Science / Engineering	Low Priority
5	100	Average (At Risk)	Strong	Improve consistency	Needs consistency	Medium Priority
6	89	High Performer	Strong	Advanced learning	Data Science / Engineering	Low Priority
7	74	Low Performer	Needs Improvement	Start with basics	Analytical roles	High Priority
8	71	Average (At Risk)	Needs Improvement	Improve consistency	Needs consistency	Medium Priority
9	85	High Performer	Strong	Advanced learning	Analytical roles	Low Priority

Download Data

Developed by **Syed Azmath & Mohammed Afnan S**

AI Learning Navigator Project

8. KEY INSIGHTS:

1. Student performance is multi-dimensional and depends on study habits, attendance, and preparation.

2. Study time has a strong positive impact on performance.

3. Absences alone are not a strong predictor but affect consistency.

4. The "Average (At Risk)" group is the most critical segment, as these students have high potential but inconsistent behavior.

4. Consistency plays a more important role than raw performance scores in long-term success.

9. AI-BASED INTERVENTION STRATEGY:

The system integrates clustering, recommendation logic, skill gap analysis, and career alignment to form a complete decision-making framework.

Instead of just analyzing data, it provides actionable insights and personalized guidance for each student, making it suitable for real-world educational applications.

10. CONCLUSION:

This project demonstrates that student performance cannot be evaluated using a single metric. By combining multiple factors such as study habits, attendance, and preparation, the system provides a comprehensive understanding of student behaviour.

The integration of clustering, personalized recommendations, skill gap analysis, and career alignment transforms raw data into meaningful insights.

The identification of the “Average (At Risk)” group highlights a key opportunity for targeted intervention, making the system practical, scalable, and impactful for modern education systems.